

INTERNATIONAL UNIVERSITY – VNU-HCMC · SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IT171IU · Statistical Learning

Task 0 – Check your environments

2.8 Task 0 – Check your environments

Run the cell below in **Google Colab** and in one **desktop tool** of your choice. Paste a screenshot of both outputs into the work area.

```
# [ADDED] worked example - run this cell
# Environment check - run in each environment and include the output in your report
import sys, platform, importlib
print("Python version :", sys.version.split()[0])
print("Executable      :", sys.executable)
print("Operating sys.  :", platform.system(), platform.release())
for pkg in ["numpy", "pandas", "matplotlib", "sklearn", "statsmodels", "ISLP"]:
    try:
        m = importlib.import_module(pkg)
        print(f"{pkg:<12} {getattr(m, '__version__', 'installed')}")
    except ImportError:
        print(f"{pkg:<12} NOT INSTALLED 📦 run the Setup cell / pip install ISLP")
```

[2] Python

```
... Python version : 3.12.5
Executable      : c:\Users\ADMIN\IT171IU-labs\venv\Scripts\python.exe
Operating sys.  : Windows 11
numpy           2.5.3
pandas          2.3.3
matplotlib      3.11.2
sklearn         1.9.1
statsmodels     0.15.0
ISLP            0.4.1
```

```
# 📌 Task 0 - environment check in two environments - your code
# Write your code below (replace the TODOs), then run the cell (Shift+Enter).
full_name = "Cao Hoang Duy" # <- replace with your full name
print(f"Hello, {full_name}! My statistical-learning environment works.")
```

[3] X student-work + Tag Python

```
... Hello, Cao Hoang Duy! My statistical-learning environment works.
```

```
!pip install ISLP
import sys, platform, importlib
print("Python version :", sys.version.split()[0])
print("Executable      :", sys.executable)
print("Operating sys.  :", platform.system(), platform.release())
for pkg in ["numpy", "pandas", "matplotlib", "sklearn", "statsmodels", "ISLP"]:
    try:
        m = importlib.import_module(pkg)
        print(f"{pkg:<12} {getattr(m, '__version__', 'installed')}")
    except ImportError:
        print(f"{pkg:<12} NOT INSTALLED - run the Setup cell / pip install ISLP")
# 📌 Task 0 - environment check in two environments - your code
# Write your code below (replace the TODOs), then run the cell (Shift+Enter).
full_name = "Cao Hoang Duy" # <- replace with your full name
print(f"Hello, {full_name}! My statistical-learning environment works.")
```

```
Python version : 3.13.15
Executable      : /usr/bin/python3
Operating sys.  : Linux 6.6.122+
numpy           2.1.3
pandas          2.2.3
matplotlib      3.10.0
sklearn         1.6.1
statsmodels     0.15.0
ISLP            0.4.1
Hello, Cao Hoang Duy! My statistical-learning environment works.
```

Exercise 1 – NumPy warm-up

Key Numbers: The shape of matrix M is (4, 5) with an ndim of 2 and dtype of int32 (or int64).

Interpretation: When creating a sub-matrix using a standard slice like $B = M[:2, :2]$, NumPy creates a "view", meaning any changes made to B will directly modify the original matrix M. Using `.copy()` creates a completely independent clone in the computer's memory, so modifying the copy leaves the original matrix unaffected.

```
... M:
[[ 2  4  6  8 10]
 [12 14 16 18 20]
 [22 24 26 28 30]
 [32 34 36 38 40]]

M.shape: (4, 5), M.ndim: 2, M.dtype: int64
Transposed M:
[[ 2 12 22 32]
 [ 4 14 24 34]
 [ 6 16 26 36]
 [ 8 18 28 38]
 [10 20 30 40]]

Second row of M:
[12 14 16 18 20]

Last column of M:
[10 20 30 40]

Sub-matrix (rows 2-3, cols 1-3):
[[12 14 16]
 [22 24 26]]

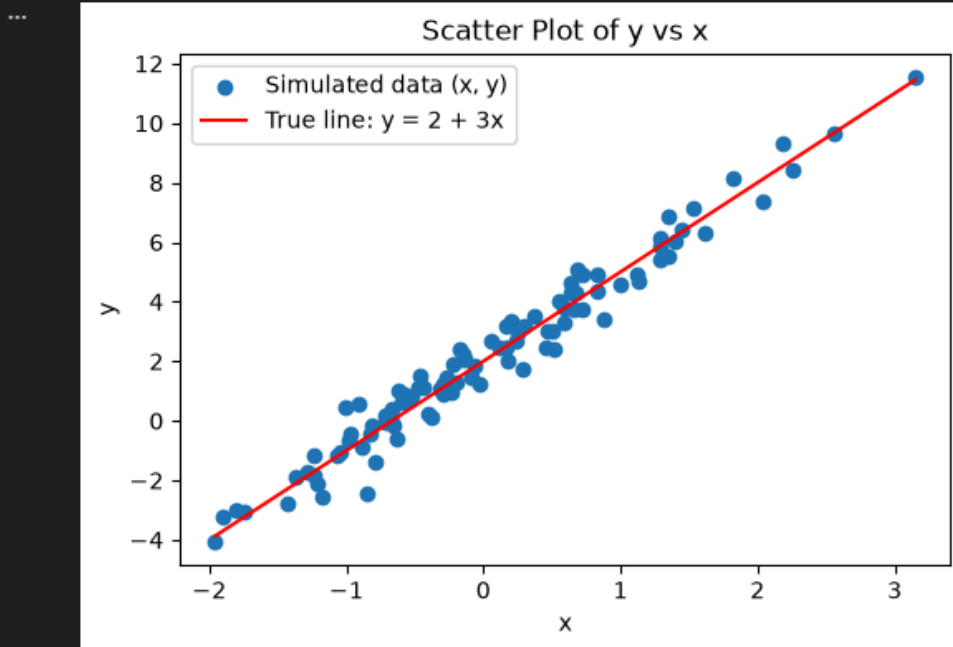
Row means: [ 6. 16. 26. 36.]
Column std devs: [11.18033989 11.18033989 11.18033989 11.18033989 11.18033989]
M[0, 0] after changing B[0, 0] to -1: -1
M[0, 0] after changing B_copy[0, 0] to -1: 2
```

Exercise 2 – Random numbers, correlation and a saved figure

Key Numbers: The correlation between x and y is approximately 0.98.

Interpretation: The output numbers and the plot remain exactly the same upon re-running the cell because the random generator was initialized with a fixed seed using `np.random.default_rng(2026)`. If the seed is removed, the function will generate completely unpredictable random values on each run, changing both the correlation output and the plot.

```
... Correlation matrix between x and y:  
[[1.      0.982033]  
 [0.982033 1.      ]]
```



Exercise 3 – Cleaning the Auto data

Key Numbers: The original data contains 5 rows with missing values (parsed as NaN), leaving exactly 392 rows in the dataset after using dropna().

Interpretation: Variables such as origin and name are qualitative, while mpg, weight, horsepower, displacement, acceleration, and year are quantitative.

```
... (a) Number of rows with missing values (NaN): 5
      Data shape after dropping: (392, 9)

      (c) Summary of quantitative variables:
```

	range	mean	std
mpg	37.6	23.445918	7.805007
cylinders	5.0	5.471939	1.705783
displacement	387.0	194.411990	104.644004
horsepower	184.0	104.469388	38.491160
weight	3527.0	2977.584184	849.402560
acceleration	16.8	15.541327	2.758864
year	12.0	75.979592	3.683737

Exercise 4 – Subsetting rows with `iloc[]`

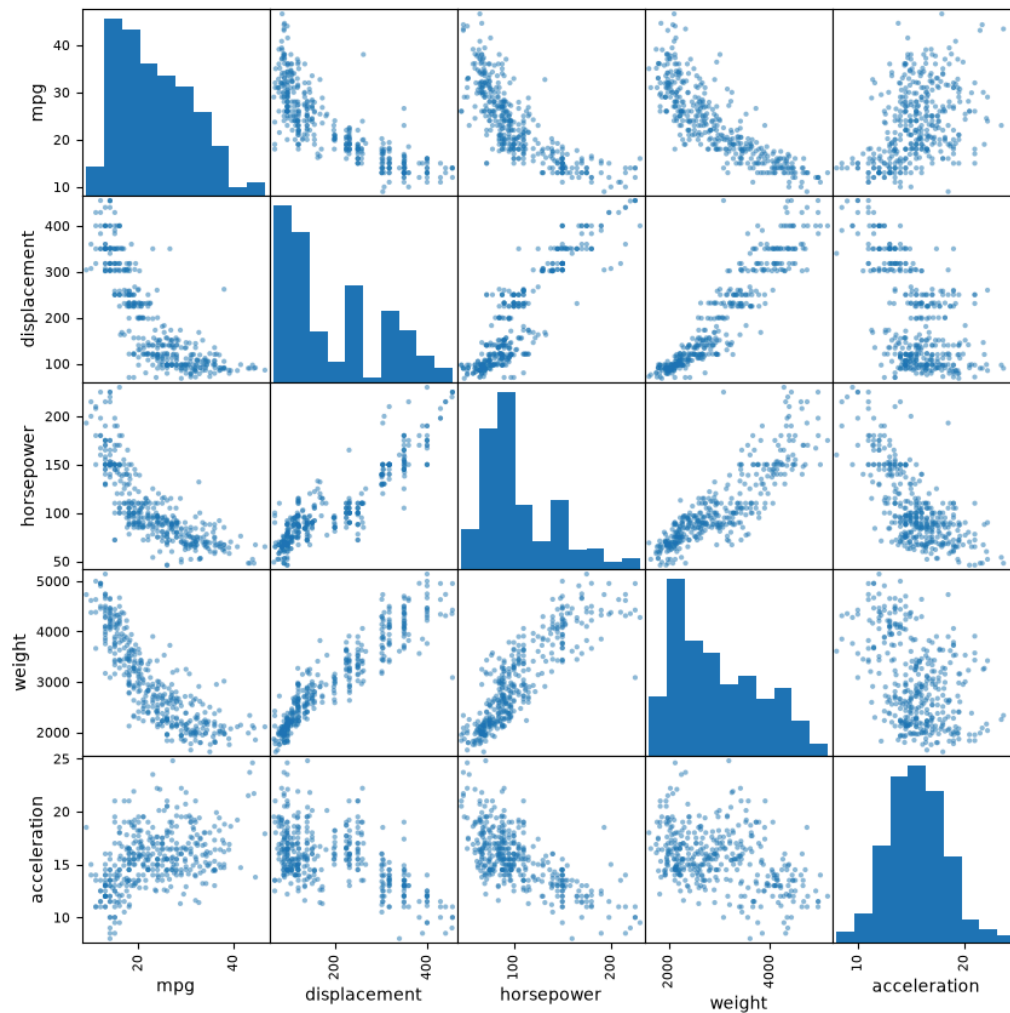
Interpretation: Removing observations 10 through 85 heavily impacts the averages and ranges of variables like mpg, weight, and displacement. Because the dataset is roughly ordered chronologically by year, this specific subset removes a concentrated cluster of older, heavier, and less fuel-efficient vehicles from the early 1970s, which artificially inflates the overall fuel efficiency averages of the remaining data.

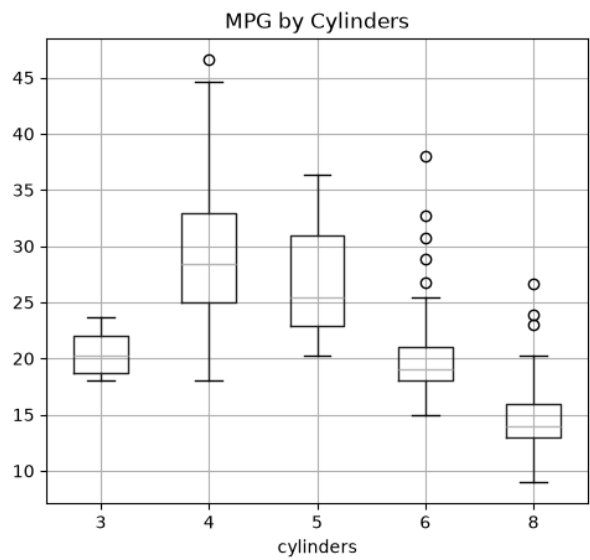
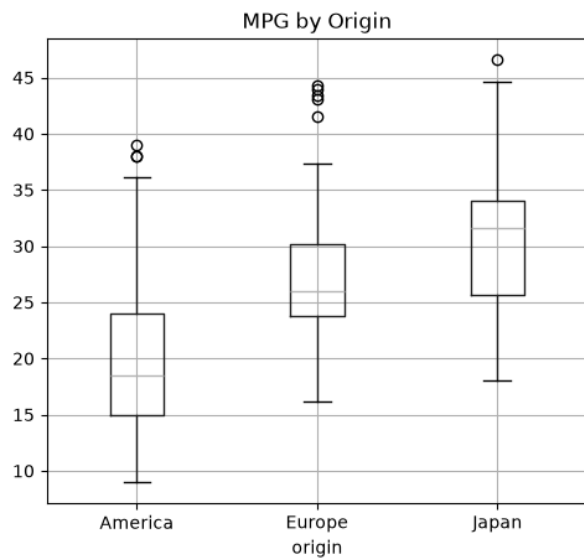
```
... Summary of quantitative variables after subsetting:
```

	range	mean	std
mpg	35.6	24.404430	7.867283
cylinders	5.0	5.373418	1.654179
displacement	387.0	187.240506	99.678367
horsepower	184.0	100.721519	35.708853
weight	3348.0	2935.971519	811.300208
acceleration	16.3	15.726899	2.693721
year	12.0	77.145570	3.106217

Exercise 5 – Graphical exploration

- Three Findings:
 1. Variables such as weight, displacement, and horsepower exhibit a strong negative correlation with mpg; as they increase, fuel efficiency decreases.
 2. The relationships between mpg and its primary numerical predictors are distinctly non-linear, often forming a convex curve rather than a straight line.
 3. Categorical variables are highly predictive: the box plots show that vehicles originating from Japan or Europe, as well as vehicles with fewer cylinders (e.g., 4), achieve significantly higher mpg than American or 8-cylinder vehicles.

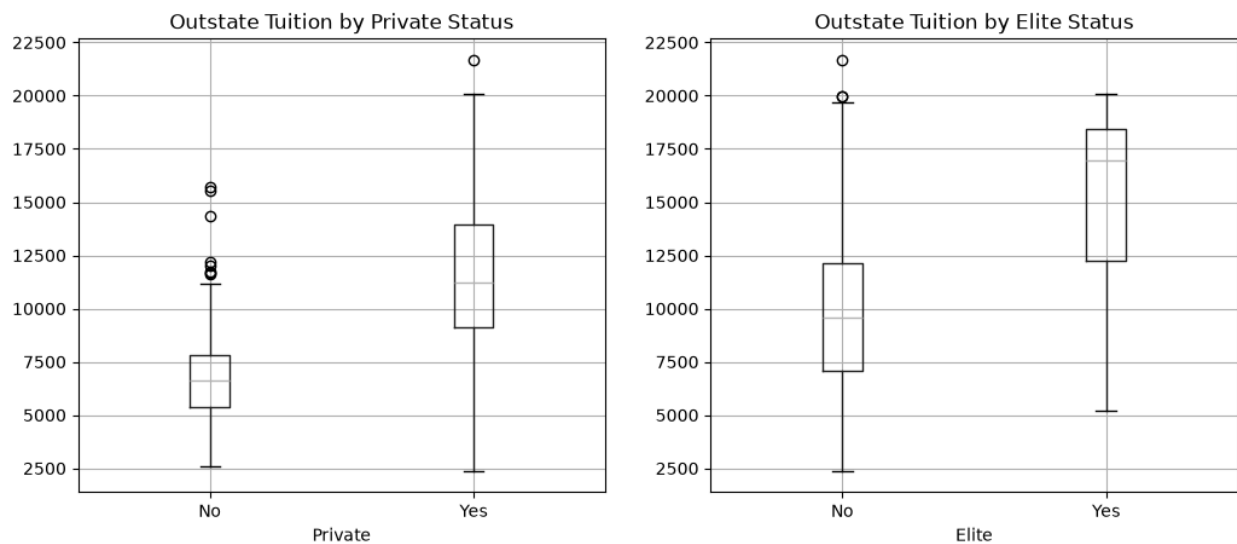


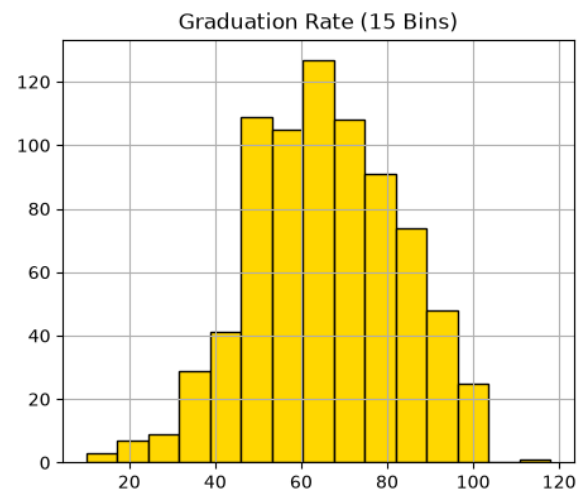
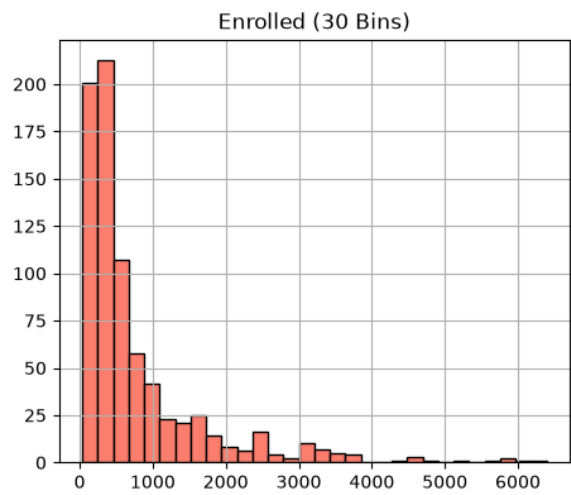
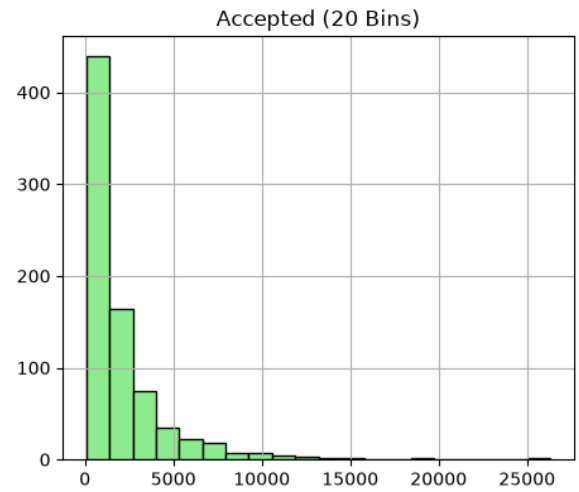
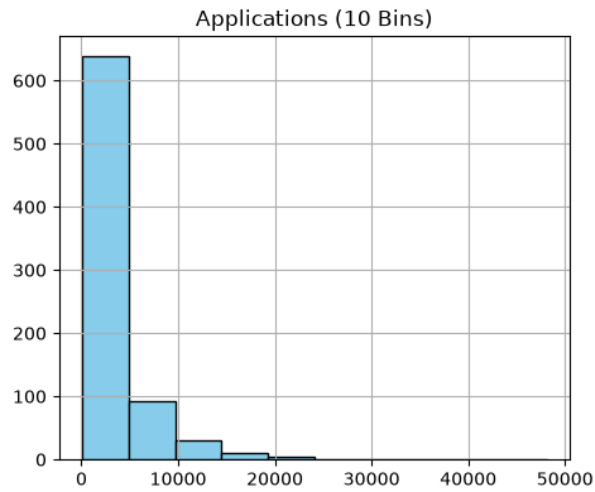


Exercise 6 – The College data set

Key Numbers: There are 565 private colleges out of 777 total institutions. After creating the Elite variable based on the top 10% high-school class metric, there are 78 elite colleges.

Interpretation: The box plots illustrate that out-of-state tuition is significantly higher at private and elite colleges compared to public or non-elite ones. The 2x2 grid of histograms reveals that application and enrollment metrics are heavily right-skewed (a few colleges receive massive application volumes), whereas graduation rates follow a much more normal, bell-shaped distribution.





Exercise 7 – Loops and string formatting

Interpretation: A for loop efficiently automates the extraction of statistics across multiple columns. When calculating the share of cars above the mean, taking the `.mean()` of a Boolean series (`Auto_clean[col] > mean_val`) automatically treats True as 1 and False as 0, elegantly yielding the mathematical proportion without needing additional counters.

```
...   mpg          mean = 23.45  sd = 7.81  above-mean share = 47.4%
      cylinders    mean = 5.47   sd = 1.71  above-mean share = 47.4%
      displacement mean = 194.41  sd = 104.64 above-mean share = 43.4%
      horsepower   mean = 104.47  sd = 38.49  above-mean share = 37.8%
      weight       mean = 2977.58 sd = 849.40 above-mean share = 43.4%
      acceleration mean = 15.54   sd = 2.76  above-mean share = 45.7%
      year         mean = 75.98   sd = 3.68  above-mean share = 54.1%
```